

B.Sc. Engg. 4th Year Odd Semester 2025

DBMS LAB

DISCUSSION-2

Use the Pubs database and explore/perform the following topics/ tasks.

1. Joining Tables [Inner join]

Discussion

The following query shows author's last name and title id of books.

```
SELECT au_lname, title_id FROM authors JOIN titleauthor ON authors.au_id=titleauthor.au_id
```

Task 1:

- i Show the title of a book, the corresponding author name.
- ii Show the title of a book, the corresponding author and publisher name.

2. The Cartesian product

The following query shows author's last name and city along with publisher's name *and* city

```
SELECT au_lname, pub_name FROM authors, publishers
```

Task 2:

- iii) Show the author name, city, publisher name **and city** only for which the authors and the publishers live in the same city

3. Nested query

```
SELECT * FROM titles WHERE royalty = (SELECT avg(royalty) FROM titles)
```

Task 3:

- iv) Show the author name(s) of the book which has the maximum royally Hint: use IN in place of =

4. Creating a table:

A table called CustomerAndSuppliers is to be created on the following schema

Field name	Data type	Size	Requirement
cust_id	character	6	1. Primary key 2. Starting with C or S and then 5 digits i.e. C00001 or S0001.
cust_fnamee	character	15	NULL nut allowed
cust_lname	character	15	Use variable character size
cust_address	text		
cust_telno	character	12	Must follow the format like 012-34567890
cust_city	character	12	Default value is Rajshahi
sales_amnt	money		Negative values not allowed
proc_amnt	money		Negative values tint allowed

The corresponding SQL statement is as follows:

CREATE TABLE CustomerAndSuppliers

(

cust_id CHAR (6) PRIMARY KEY CHECK (cust_id LIKE '[CS] [0-9][0-9] [0-9] [0-9] [0-9] '),

cust_fname CHAR(15) NOT NULL,

cust_fname VARCHAR (15),

cust_address TEXT,

cust_telno(12) CHECK (cust_telno LIKE '[0-9][0-9] [0-9]-[0-9] [0-9] [0-9] [0-9] [0-9] [0-9] [0-9] [0-9] [0-9] '),

cust_city CHAR (12) DEFAULT 'Rajshahi',

sales_amnt MONEY CHECK (sales_amnt>=0),

proc_amnt MONEY CHECK (proc_amnt>=0)

)

5. Inserting data into a table:

Example:

INSERT CustomerAndSuppliers (cust_id, cust_fname, cust_lname, cust_address, cust_telno, cust_city, sales_amnt, proc_amnt) VALUES ('C00001', 'Iqbal', '117/3 Motihar', '017-00000001', 'Rajshahi', 0, 0)

Task4: Create the following Table

Table name: Item

Field name	Data type	Size	Requirement
item_id	Character	6	1. Primary key 2. Starting with P and then 5 numbers i.e. A0001.
item_name	character	12	
item_category	character	10	Example: Electrical, Mechanical, Software, Books etc.
item_price	float	12	Negative values not allowed
item_qoh	integer		Negative values not allowed
item_last_sold	date		Default value is current date

Table: Transactions

Field name	Data type	Size	Requirement
tran_id	Character	10	1. Primary key 2. Starting with T and then 9 numbers i.e. T000000001.
item_id	character	6	Foreign key with reference to item table
cust_id	character	5	Foreign key with reference to customer table
tran_type	character	1	Either S or O, (S for sales Order to supplier)
tran_quantity	integer	12	Only positive values.
tran_date	date and time		Default is current date